

DIFFERENTIAL EQUATIONS LECTURE NOTES

WEEK

IX

HIGHER ORDER ODES

$$P_n(t)y^{(n)} + P_{n-1}(t)y^{(n-1)} + \dots + P_1(t)y' + P_0(t)y = G(t) \rightarrow \text{nth order linear ode's}$$

$\nexists P_n(t) \neq 0$

If $P_n(t) \neq 0 \quad \forall t \in I$ then divide DE with $P_n(t)$.

$$y^{(n)} + \frac{P_{n-1}(t)}{P_n(t)} y^{(n-1)} + \dots + \frac{P_1(t)}{P_n(t)} y' + \frac{P_0(t)}{P_n(t)} y = \frac{G(t)}{P_n(t)}$$

OR

$$y^{(n)} + a_{n-1}(t)y^{(n-1)} + \dots + a_1(t)y' + a_0(t)y = f(t) \text{ "standard form" of an nth order linear ODE}$$

Definition If $f(t) = 0$ then DE is called **homogeneous DE (H)**
If $f(t) \neq 0$ then DE is called **nonhomogeneous DE (NH)**

Then,

$$y^{(n)} + a_{n-1}(t)y^{(n-1)} + \dots + a_1(t)y' + a_0(t)y = f(t) \quad (N)$$
$$y^{(n)} + a_{n-1}(t)y^{(n-1)} + \dots + a_1(t)y' + a_0(t)y = 0$$

Notation If $a_i(t), i=0,1,\dots,n-1$ are continuous functions on some interval I , then we define the Differential Operator L by the equation

$$L[y] = y^{(n)} + a_{n-1}(t)y^{(n-1)} + \dots + a_1(t)y' + a_0(t)y$$

where the operator $L = D^n + a_{n-1}D^{n-1} + \dots + a_1(t)D + a_0(t)$

with $D^n = \frac{d^n}{dt^n}$

①

Ex 7 $L = D^2 + D - 1 = \frac{d^2}{dt^2} + \frac{d}{dt} - 1$ $L[y] = y'' + y' - y$

so $L[y] = f(t) \quad (N)$, $L[y] = 0 \quad (H)$.

where $L = D^n + a_{n-1}(t)D^{n-1} + \dots + a_1(t)D + a_0(t)$.

Remark The letter (L) is chosen to remind us that the operator L is linear in the sense that

$$L[\alpha y_1 + \beta y_2] = \alpha L[y_1] + \beta L[y_2] \quad \alpha, \beta: \text{constant}$$

Ex 7 $L = D^2 + D - 1$ linear map.

$$\begin{aligned} L[\alpha y_1 + \beta y_2] &= D^2[\alpha y_1 + \beta y_2] + D[\alpha y_1 + \beta y_2] - [\alpha y_1 + \beta y_2] \\ &= \alpha y_1'' + \beta y_2'' + \alpha y_1' + \beta y_2' - \alpha y_1 - \beta y_2 \\ &= \alpha \underbrace{[y_1'' + y_1' - y_1]}_{L[y_1]} + \beta \underbrace{[y_2'' + y_2' - y_2]}_{L[y_2]} \end{aligned}$$

Ex 7 Consider 2nd order linear D.E.

$$y'' + a_1(t)y' + a_0(t)y = f(t) \Rightarrow L[y] = f(t)$$

$$L = D^2 + a_1(t)D + a_0(t)$$

Ex 7 $y''' - 2y'' + y' - y = e^{-t}$

$$L[y] = \underbrace{e^{-t}}_{f(t)} \quad \text{where} \quad L = D^3 - 2D^2 + D - 1$$

Defn Define $L[y] = y^{(n)} + a_{n-1}(t)y^{(n-1)} + \dots + a_1(t)y' + a_0(t)y$

where $a_i(t)$ are cont. on some interval I . Then,

① (N) $L[y] = f(t)$ and $f(t)$ is cont. on I

② (H) $L[y] = 0$

Now, give initial conditions

② $y(t_0) = y_0, y'(t_0) = y_0', y''(t_0) = y_0'', \dots, y^{(n-1)}(t_0) = y_0^{(n-1)}$

where $y_0, y_0', \dots, y_0^{(n-1)}$ are given values

Then ① + ② is called an Initial Value Problem (I.V.P.)

Theorem Consider the IVP $y^{(n)} + a_{n-1}(t)y^{(n-1)} + \dots + a_1(t)y' + a_0(t)y = f(t)$
 $y(t_0) = y_0, y'(t_0) = y_0', \dots, y^{(n-1)}(t_0) = y_0^{(n-1)}$

where $q_i(t), i=0, 1, 2, \dots, n-1$ and $g(t)$ are continuous on an open interval I containing t_0 , then there exists exactly one solution

$y = \Phi(t)$ of the I.V.P satisfying D.E and I.C's.

Ex $3y'' + x^2y'' + 4x^3y' + 8y = e^x$ I.
 $y(1) = 3 \quad y'(1) = 6 \quad y''(1) = -3$ (---|---)

1 near 3rd degree D.E.

$$y''' + \underbrace{\frac{x^2}{3}}_{q_2(x)} y'' + \underbrace{\frac{4x^2}{3}}_{q_1(x)} y' + \underbrace{\frac{8}{3}}_{q_0} y = \underbrace{e^x}_{f(x)}$$

$g_p(x), y(x)$ are all continuous $\forall p=0,1,2 \quad \forall x \in \mathbb{R}$.

So, interval of solutions in $R = (-\infty, \infty)$ by $\exists$ P.D Theorem.

Ex Determine the intervals for which the unique solution exists for the I.C's $y(x_0) = y_0, y'(x_0) = y'_0, y''(x_0) = y''_0$

(9) $x(x-1)y'' + 3xy' + 4y = 2$ 2nd order linear. III
 $y'' + \frac{3x}{x(x-1)} y' + \frac{4}{x(x-1)} y = \frac{2}{x(x-1)}$
 I = (-∞, 0) II = (0, 1) III = (1, ∞)

 A horizontal number line with tick marks at 0 and 1. Above the line, interval I is labeled as (-∞, 0), interval II as (0, 1), and interval III as (1, ∞). Arrows point from each label towards its respective interval.

a_1, a_0 and $y(x)$ are cont. $\forall x \in \mathbb{R} - \{0, 1\}$

By 8.11 Theorem If $x_0 \in I = (-\infty, 0) \Rightarrow$ I.V.P has a unique sol. on I
 $\neq$ If $x_0 \in II = (0, 1) \Rightarrow$ I.V.P has " " " " II
 $\neq$ If $x_0 \in III = (1, \infty) \Rightarrow$ " " " " III

(b) $e^x y'' + xy' + y = \tan x$

$$y'' + \left(\frac{x}{ex}\right)y' + \left(\frac{14}{ex}\right)y = \left(\frac{100x}{ex}\right)g(x)$$

$$a_1(x), a_0(x), p(x) \text{ are cont. } \forall x \in \mathbb{R}$$

$$g(x) = \frac{\sin x}{\cos x} \cdot \frac{1}{\cos x} \quad \forall x \text{ where}$$

$$\cos x \neq 0 \Rightarrow x \neq (2k+1)\pi/2$$

$$k = 0, \pm 1, \pm 2, \dots$$

$$-5\pi/2 \quad -3\pi/2 \quad -\pi/2 \quad \pi/2 \quad 3\pi/2 \quad 5\pi/2$$

$$\text{If } x_0 \in ((2k+1)\pi/2, (2k+3)\pi/2) \\ k=0, \pm 1, \pm 2, \dots$$

$\exists$ the solution exists and unique
in that interval by $\exists! \Rightarrow$ There

$$\text{If } x_0 = 2\pi \Rightarrow I = (\frac{3\pi}{2}, \frac{5\pi}{2}) \ni 2\pi$$

$$\text{If } x_0 = 3\pi \Rightarrow I = (\frac{5\pi}{2}, \frac{7\pi}{2}) \ni 3\pi$$

Thm (Superposition Principle)

If the functions $y_1, y_2, \dots, y_n$ are solutions of

$$L[y] = y^{(n)} + a_{n-1}(t)y^{(n-1)} + \dots + a_1(t)y' + a_0(t)y = 0 \quad (H)$$

then any linear combination of these solutions

$$y = c_1 y_1 + c_2 y_2 + \dots + c_n y_n; \quad c_i \text{ constants}$$

is a solution for (H)

Proof If $y_1, y_2, \dots, y_n$ are solutions of $L[y] = 0 \Rightarrow L[y_i] = 0$

$$L[c_1 y_1 + c_2 y_2 + \dots + c_n y_n] = c_1 L[y_1] + c_2 L[y_2] + \dots + c_n L[y_n] = 0$$

$\downarrow$ by linearity of L

Question Is there any solution which can't be obtained from

$$y = c_1 y_1 + c_2 y_2 + \dots + c_n y_n$$

where $y_1, y_2, \dots, y_n$ are solutions of $L[y] = 0 \quad (H)$

Answer No, if $y_1, y_2, \dots, y_n$ are linearly independent. In other

words, if $\exists$ a point to where the

$$\text{Wronskian: } W(y_1, y_2, \dots, y_n)(t) = \begin{vmatrix} y_1 & y_2 & \dots & y_n \\ y_1' & y_2' & \dots & y_n' \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \dots & y_n^{(n-1)} \end{vmatrix} \neq 0$$

Then the functions $y_1, y_2, \dots, y_n$ are linearly independent

Definition Then, the family of solutions $y = c_1 y_1 + c_2 y_2 + \dots + c_n y_n$ with arbitrary $c_i, i=1, 2, \dots, n$, includes every solutions to (H). Thus, it is called a general solution. And the linearly independent set $\{y_1, y_2, \dots, y_n\}$ is called fundamental set of solutions of (H).

Theorem Let $\{y_1(t), y_2(t), \dots, y_n(t)\}$ be set of solutions for
(H) $L[y] = y^{(n)} + a_{n-1}(t)y^{(n-1)} + \dots + a_1(t)y' + a_0(t)y = 0$

Then, the followings are equivalent

- ① $\{y_1, y_2, \dots, y_n\}$ is linearly independent. (That is $c_1 y_1 + \dots + c_n y_n = 0 \Rightarrow c_i = 0 \forall i$)
- ② $w(y_1, y_2, \dots, y_n)(t) \neq 0 \quad \forall t \in I$
- ③ $w(y_1, y_2, \dots, y_n)(t) \neq 0$ for some $t_0 \in I$
- ④ $y = c_1 y_1 + c_2 y_2 + \dots + c_n y_n$ is general solution of (H).
- ⑤ For nonhomogeneous system $L[y] = p(t) \quad (N)$

$$y(t) = y_h + y_p$$

$\downarrow$ $\rightarrow$ particular solution of (N)
 homogeneous
 (complementary solution for (H))

- ⑥ If $\phi_1(t)$ and $\phi_2(t)$ are solutions for (N), then $y = \phi_1(t) - \phi_2(t)$ is a solution for (H).

Proof Since ϕ_1 and ϕ_2 are solutions of (N), then

$$L[\phi_1] = p(t), \quad L[\phi_2] = p(t)$$

$$L[\phi_1 - \phi_2] = \underbrace{L(\phi_1) - L(\phi_2)}_{\text{by linearity}} = p(t) - p(t) = 0$$

$$L'[\phi_1 - \phi_2] = 0 \Rightarrow \phi_1 - \phi_2 \text{ is a solution of } L[y] = 0$$

SOLUTION TECHNIQUES FOR n th ORDER LINEAR DEs.

$$y^{(n)} + a_{n-1}(t)y^{(n-1)} + \dots + a_1(t)y' + a_0(t)y = f(t) \quad (N)$$

$$y^{(n)} + a_{n-1}(t)y^{(n-1)} + \dots + a_1(t)y' + a_0(t)y = 0 \quad (H)$$

Now, consider first nonhomog and hom DE with constant

coefficients, that is $a_i(t) = a_i$ constants $\forall i = 0, 1, \dots, n-1$

$$\begin{array}{l} y^{(n)} + a_{n-1}y^{(n-1)} + \dots + a_1y' + a_0y = f(t) \quad (\tilde{N}) \\ \hline y^{(n)} + a_{n-1}y^{(n-1)} + \dots + a_1y' + a_0y = 0 \quad (\tilde{H}) \end{array}$$

First focus on $(\tilde{H})$:

$(\tilde{H})$ can be converted into a system of 1st order ODEs:

Then's

$$\begin{aligned} x_1 &= y \\ x_2 &= y' \\ x_3 &= y'' \\ &\vdots \\ x_n &= y^{(n-1)} \end{aligned}$$

differentiate
 $\longrightarrow$

$$\begin{aligned} x_1' &= y' = x_2 \\ x_2' &= y'' = x_3 \\ x_3' &= y''' = x_4 \\ &\vdots \end{aligned}$$

$$x_n' = y^{(n)} = -a_{n-1}y^{(n-1)} - a_{n-2}y^{(n-2)} - \dots - a_1y' - a_0y$$

$\downarrow$ $\parallel$ $\parallel$ $\parallel$
 $\text{by } (\tilde{H})$ x_n x_{n-1} x_2

$$x_n' = -a_{n-1}x_n - a_{n-2}x_{n-1} - \dots - a_1x_2 - a_0x_1$$

Then,

$$\begin{bmatrix} x_1' \\ x_2' \\ \vdots \\ x_n' \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \dots & \dots & 1 \\ -a_0 & -a_1 & \dots & -a_{n-2} & -a_{n-1} \end{bmatrix}}_A \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

We now know how to solve the system of 1st order linear equations with constant coefficient matrix A .

So, look for characteristic equation.

$\det(A - \lambda I) = 0$ to find eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$. Then, obtain corresponding eigenvectors $\vec{v}^{(1)}, \vec{v}^{(2)}, \dots, \vec{v}^{(n)}$

So, the equation $\vec{x}^{(1)} = e^{\lambda_1 t} \vec{v}^{(1)}, \vec{x}^{(2)} = e^{\lambda_2 t} \vec{v}^{(2)}, \dots, \vec{x}^{(n)} = e^{\lambda_n t} \vec{v}^{(n)}$

Now, calculate

$$\det(A - \lambda I) = \begin{vmatrix} -\lambda & 1 & 0 & \dots & 0 \\ 0 & -\lambda & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ -a_0 & -a_1 & \dots & -a_{n-2} & 1 & -\lambda - a_{n-1} \end{vmatrix}$$

$$= (-\lambda)^{n-1} \begin{vmatrix} -\lambda & 1 & \dots & 0 \\ 0 & -\lambda & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -a_1 & -a_2 & \dots & -\lambda - a_{n-1} \end{vmatrix} + 0 + 0 + \dots + (-1)^{n-1} a_0 \begin{vmatrix} 1 & 0 & \dots & 0 \\ 0 & -\lambda & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & \dots & 0 & 1 & -\lambda \end{vmatrix}$$

Use induction and it follows from that

$$\det(A - \lambda I) = (-1)^n (\lambda^n + a_{n-1} \lambda^{n-1} + a_{n-2} \lambda^{n-2} + \dots + a_1 \lambda + a_0) = 0$$

$$\boxed{\lambda^n + a_{n-1} \lambda^{n-1} + a_{n-2} \lambda^{n-2} + \dots + a_1 \lambda + a_0 = 0} \rightarrow \text{Char eq.}$$

Remark Since $y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = 0$, it is observed that one can obtain the char equation by putting λ instead of y in (H) .

$\Rightarrow y'' + a_1 y' + a_0 y = 0$

$$\begin{aligned} x_1 &= y \\ x_2 &= y' \end{aligned} \quad \xrightarrow{\text{diff}} \quad \begin{aligned} x_1' &= y' = x_2 \\ x_2' &= y'' = -a_1 y' - a_0 y = -a_1 x_2 - a_0 x_1 \end{aligned}$$

by DE

$$\begin{bmatrix} x_1' \\ x_2' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -a_0 & -a_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\det(A - \lambda I) = \begin{vmatrix} -\lambda & 1 \\ -a_0 & -\lambda - a_1 \end{vmatrix} = \lambda^2 + a_1 \lambda + a_0 = (-1)^2 (\lambda^2 + a_1 \lambda + a_0) = 0$$

$$\boxed{\lambda^2 + a_1 \lambda + a_0 = 0}$$

(7)

Ex $y''' + a_2 y'' + a_1 y' + a_0 y = 0$

$$\begin{aligned} x_1 &= y \\ x_2 &= y' \\ x_3 &= y'' \end{aligned} \quad \Rightarrow \quad \begin{aligned} x_1' &= y' = x_2 \\ x_2' &= y'' = x_3 \\ x_3' &= y''' = -a_2 x_3 - a_1 x_2 - a_0 x_1 \end{aligned}$$

$$\begin{bmatrix} x_1' \\ x_2' \\ x_3' \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\det(A - \lambda I) = \begin{vmatrix} -\lambda & 1 & 0 \\ 0 & -\lambda & 1 \\ -a_0 & -a_1 & -\lambda - a_2 \end{vmatrix} = (-\lambda)(\lambda^2 + a_2\lambda + a_1) - a_0 \cdot 1$$

$$= (-1)^3 (\lambda^3 + a_2\lambda^2 + a_1\lambda + a_0) = 0$$

Remark Once we find the roots of characteristic eqn. we know how to solve the system.
We have 3 cases for roots of char eqn. $\det(A - \lambda I) = 0$

Case I: REAL, DISTINCT ROOTS FOR CHAR. EQ.

$$y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = 0$$

$$\begin{aligned} x_1 &= y \\ x_2 &= y' \\ &\vdots \\ x_n &= y^{(n-1)} \end{aligned} \quad \Rightarrow \quad \vec{x}' = A \vec{x} \quad \text{where} \quad A = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ -a_0 & -a_1 & \dots & \dots & -a_{n-1} \end{bmatrix}$$

Char. eqn. $\det(A - \lambda I) = \lambda^n + a_{n-1} \lambda^{n-1} + \dots + a_1 \lambda + a_0 = 0$

If roots are real & distinct then A is diagonalizable that is

$\exists$ a matrix P such that $P^{-1}AP = J = \begin{bmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \ddots \\ & & & \lambda_n \end{bmatrix}$

$P = [\vec{v}^{(1)} \ \vec{v}^{(2)} \ \dots \ \vec{v}^{(n)}]$ then the solution of system is

$$\begin{aligned} \vec{x}_{\text{general}} &= \Psi(t) \vec{c} = P e^{Jt} \vec{c} = \begin{bmatrix} \vec{v}^{(1)}(t) & \vec{v}^{(2)}(t) & \dots & \vec{v}^{(n)}(t) \end{bmatrix} \begin{bmatrix} e^{\lambda_1 t} & 0 & \dots & 0 \\ 0 & e^{\lambda_2 t} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & e^{\lambda_n t} \end{bmatrix} \vec{c} \\ &= \begin{bmatrix} \vec{v}^{(1)}(t) e^{\lambda_1 t} & \vec{v}^{(2)}(t) e^{\lambda_2 t} & \dots & \vec{v}^{(n)}(t) e^{\lambda_n t} \end{bmatrix} \vec{c} \end{aligned}$$

$$= c_1 \vec{x}_1(t) + c_2 \vec{x}_2(t) + \dots + c_n \vec{x}_n(t)$$

In this case $\vec{x}_i = \begin{bmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{in} \end{bmatrix}$

$$\begin{aligned} x_1 &= y \\ x_2 &= y' \\ &\vdots \\ x_n &= y^{(n-1)} \end{aligned}$$

$$\Rightarrow \begin{aligned} x_{i1} &= y \\ x_{i2} &= y' \\ &\vdots \\ x_{in} &= y^{(n-1)} \end{aligned}$$

Thus, $\vec{x}_i = \begin{bmatrix} y_i \\ y_i' \\ \vdots \\ y_i^{(n-1)} \end{bmatrix}$ are free solutions of $\vec{x}' = A\vec{x}$

$\Psi(t) = [\vec{x}_1 \ \vec{x}_2 \ \dots \ \vec{x}_n] = \begin{bmatrix} y_1 & y_2 & \dots & y_n \\ y_1' & y_2' & \dots & y_n' \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \dots & y_n^{(n-1)} \end{bmatrix}$ For solution of 1th order ODE first row of the fundamental matrix

Remark

We know that

$$\det \Psi(t) = \begin{vmatrix} \vec{x}_1 & \vec{x}_2 & \dots & \vec{x}_n \\ x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{nn} \end{vmatrix} = \begin{vmatrix} y_1 & y_2 & \dots & y_n \\ y_1' & y_2' & \dots & y_n' \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \dots & y_n^{(n-1)} \end{vmatrix} \neq 0$$

for the system

$n(y_1, y_2, \dots, y_n)$

So, nonvanishing of $y_1, y_2, \dots, y_n \neq 0 \Rightarrow \{y_1, y_2, \dots, y_n\}$ is a linearly independent set, so it constitutes a set of fundamental solutions for higher order DE, $y^{(n)} + a_{n-1}y^{(n-1)} + \dots + a_0y = 0(t)$

So, general solution $y_g = c_1y_1 + c_2y_2 + \dots + c_ny_n$

Since

$$\Psi(t) = \begin{bmatrix} v_{11}e^{\lambda_1 t} & v_{12}e^{\lambda_2 t} & \dots & v_{1n}e^{\lambda_n t} \\ v_{21}e^{\lambda_1 t} & v_{22}e^{\lambda_2 t} & \dots & v_{2n}e^{\lambda_n t} \\ \vdots & \vdots & \ddots & \vdots \\ v_{m1}e^{\lambda_1 t} & v_{m2}e^{\lambda_2 t} & \dots & v_{mn}e^{\lambda_n t} \end{bmatrix}$$

$$\{y_1 = v_{11}e^{\lambda_1 t}, y_2 = v_{12}e^{\lambda_2 t}, \dots, y_n = v_{1n}e^{\lambda_n t}\}$$

constants

So $\{y_1 = e^{\lambda_1 t}, y_2 = e^{\lambda_2 t}, \dots, y_n = e^{\lambda_n t}\}$ is the fundamental set of solutions

Ex $y'' - 2y' - 3y = 0$ solve

1st way Use system.

$$\begin{aligned} x_1 &= y & \frac{d}{dt} & \Rightarrow & x_1 = y' = x_2 \\ x_2 &= y' & & & x_2 = y'' = 2x_2 + 3x_1 \end{aligned}$$

$$\Rightarrow \begin{bmatrix} x_1' \\ x_2' \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 \\ 3 & 2 \end{bmatrix}}_A \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\det(A - \lambda D) = \begin{vmatrix} -\lambda & 1 \\ 3 & 2-\lambda \end{vmatrix} = \lambda^2 - 2\lambda - 3 = 0 \quad (\lambda-3)(\lambda+1) = 0$$

$$\lambda = 3 \quad \lambda = -1$$

real distinct

$$\underline{\lambda=3} \quad \begin{bmatrix} -3 & 1 \\ 3 & -1 \end{bmatrix} \longrightarrow \begin{bmatrix} 3 & -1 \\ 0 & 0 \end{bmatrix} \quad 3v_1 - v_2 = 0 \quad 3v_1 = v_2$$

$$v_{\lambda=3} = \text{span} \left\{ \begin{bmatrix} 1 \\ 3 \end{bmatrix} \right\}$$

$$\vec{x}_1(t) = e^{3t} \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$\underline{\lambda=-1} \quad \begin{bmatrix} 1 & 1 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

$$v_1 + v_2 = 0 \quad v_{\lambda=-1} = \text{span} \left\{ \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right\}$$

$$\vec{x}_2(t) = e^{-t} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$v(t) = [\vec{x}_1 \quad \vec{x}_2] = \begin{bmatrix} \underbrace{e^{3t}}_{y_1} & \underbrace{e^{-t}}_{y_2} \\ \underbrace{3e^{3t}}_{y_1'} & \underbrace{-e^{-t}}_{y_2'} \end{bmatrix}$$

$$\text{So } y_{\text{general}}(t) = c_1 y_1(t) + c_2 y_2(t) = c_1 e^{3t} - c_2 e^{-t} = c_1 e^{3t} + c_3 e^{-t}$$

2nd way

write char. equation

$$y'' - 2y' - 3y = 0$$

$$\lambda^2 - 2\lambda - 3 = 0$$

$$(\lambda-3)(\lambda+1) = 0$$

$$\lambda = 3 \quad \lambda = -1 \quad \text{real distinct}$$

$$y_{\text{general}} = c_1 e^{3t} + c_2 e^{-t}$$

CASE II COMPLEX ROOTS FOR CNR EQ. 11

If $\lambda_1 = \alpha + i\beta \Rightarrow$ complex eigenvalue (root) then $\lambda_2 = \bar{\lambda}_2 = \alpha - i\beta$
 is also an eigenvalue and λ_1 and λ_2 produce solutions of the
 form $\{e^{\alpha t} \cos \beta t, e^{\alpha t} \sin \beta t\}$ which are real solutions

By Case I The solutions are, since λ_1 and λ_2 are distinct

$$\vec{\phi}_1(t) = e^{\lambda_1 t} = e^{(\alpha + i\beta)t} = e^{\alpha t} e^{i\beta t} = e^{\alpha t} (\cos(\beta t) + i \sin(\beta t)) \text{ "complex"}$$

$$\vec{\phi}_2(t) = e^{\lambda_2 t} = e^{(\alpha - i\beta)t} = e^{\alpha t} e^{-i\beta t} = e^{\alpha t} (\cos(-\beta t) + i \sin(-\beta t))$$

$\underbrace{\cos(-\beta t)}_{\cos \beta t} \quad \underbrace{i \sin(-\beta t)}_{-i \sin \beta t}$

$$\vec{x}_1 = \frac{\vec{\phi}_1 + \vec{\phi}_2}{2} = e^{\alpha t} \cos \beta t \text{ "real sol"}$$

$\underbrace{\text{Re}(e^{\lambda_1 t} \vec{v})}$

$$\vec{x}_2 = \frac{\vec{\phi}_1 - \vec{\phi}_2}{2i} = e^{\alpha t} \sin \beta t \text{ "real sol"}$$

$\underbrace{\text{Im}(e^{\lambda_1 t} \vec{v})}$

Then,

$$y_p = C_1 e^{\alpha t} \cos \beta t + C_2 e^{\alpha t} \sin \beta t$$

$$e^{\text{Re}(\lambda)t} \cos(\text{Im}(\lambda)t)$$

$$e^{\text{Re}(\lambda)t} \sin(\text{Im}(\lambda)t)$$

Ex $y'' + 4y = 0$ solve

1st way Use systems

$$\begin{aligned} x_1 &= y \\ x_2 &= y' \end{aligned} \Rightarrow \begin{aligned} x_1' &= y' = x_2 \\ x_2' &= y'' = -4x_1 \end{aligned} \Rightarrow \begin{bmatrix} x_1' \\ x_2' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -4 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

A.

Char eq. $\det(A - \lambda I) = \begin{vmatrix} -\lambda & 1 \\ -4 & -\lambda \end{vmatrix} = \lambda^2 + 4 = 0 \Rightarrow \lambda = \pm 2i$

$\text{Re}(\lambda) = 0$

$\lambda = 2i$

$$\begin{bmatrix} -2i & 1 \\ -4 & -2i \end{bmatrix} \rightarrow \begin{bmatrix} -2i & 1 \\ 0 & 0 \end{bmatrix} \quad -2iv_1 + v_2 = 0 \quad \vec{v} = \begin{bmatrix} 1 \\ 2i \end{bmatrix}$$

$\text{Im}(\lambda) = 2$

$$e^{\lambda t} \vec{v} = e^{2it} \begin{bmatrix} 1 \\ 2i \end{bmatrix} = (\cos 2t + i \sin 2t) \begin{bmatrix} 1 \\ 2i \end{bmatrix} = \begin{bmatrix} \cos 2t \\ -2 \sin 2t \end{bmatrix} + i \begin{bmatrix} \sin 2t \\ 2 \cos 2t \end{bmatrix}$$

$\underbrace{\begin{bmatrix} \cos 2t \\ -2 \sin 2t \end{bmatrix}}_{\vec{x}_1} \quad \underbrace{\begin{bmatrix} \sin 2t \\ 2 \cos 2t \end{bmatrix}}_{\vec{x}_2}$

$$\gamma(t) = [\vec{x}_1(t) \quad \vec{x}_2(t)] = \begin{bmatrix} \cos 2t & \sin 2t \\ -2 \sin 2t & 2 \cos 2t \end{bmatrix}$$

(d)

$y_p = C_1 y_1 + C_2 y_2 = C_1 \cos 2t + C_2 \sin 2t$

2nd way $y'' + 4y = 0 \Rightarrow$ char eq. $\lambda^2 + 4 = 0 \quad \lambda_{1,2} = \pm 2i$

$$y_1 = e^{\operatorname{Re}(\lambda)t} \cos(\operatorname{Im}(\lambda)t) = e^{0t} \cos 2t = \cos 2t.$$

$$\operatorname{Re}(\lambda) = 0$$

$$\operatorname{Im}(\lambda) = 2$$

$$y_2 = e^{\operatorname{Re}(\lambda)t} \sin(\operatorname{Im}(\lambda)t) = e^{0t} \sin 2t = \sin 2t.$$

$$y_g = C_1 \cos 2t + C_2 \sin 2t$$

$$\sin(-2t) = -\sin 2t$$

Exm $y'' + y' + y = 0$ Solve

$$\text{char eq. } \lambda^2 + \lambda + 1 = 0 \Rightarrow \lambda_{1,2} = \frac{-1 \pm \sqrt{1-4}}{2} = -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$$

$$\lambda = -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i \Rightarrow \operatorname{Re} \lambda = -\frac{1}{2} \quad \operatorname{Im} \lambda = \frac{\sqrt{3}}{2}$$

$$y_1 = e^{-\frac{1}{2}t} \cos\left(\frac{\sqrt{3}}{2}t\right), \quad y_2 = e^{-\frac{1}{2}t} \sin\left(\frac{\sqrt{3}}{2}t\right)$$

$$y_g = C_1 y_1 + C_2 y_2.$$

Case III Repeated Roots for Char. Equation

Thm Let λ be a root of Char equation.

$$\det(A - \lambda I) = (-1)^n (\lambda^n + a_{n-1}\lambda^{n-1} + \dots + a_1\lambda + a_0) = 0$$

with $g(\lambda) = k$. Then, there has to be only one Jordan block i.e. $J = \begin{bmatrix} \lambda & 1 & 0 & \dots & 0 \\ 0 & \lambda & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \dots & \lambda \end{bmatrix}$ for the solution of the system

$$\vec{X}' = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \dots & 1 \\ -a_0 & -a_1 & \dots & \dots & -a_{n-1} \end{bmatrix} \vec{X} \quad \text{which is obtained from a higher order ODE}$$

$$y^{(n)} + a_{n-1}y^{(n-1)} + \dots + a_1y' + a_0y = 0$$

Here $J_\lambda = \begin{bmatrix} \lambda & 1 & 0 & \dots & 0 \\ 0 & \lambda & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \dots & \lambda \end{bmatrix}$ and $e^{J_\lambda t} = e^{\lambda t} \begin{bmatrix} 1 & t & t^2/2! & \dots & \frac{t^{k-1}}{(k-1)!} \\ 0 & 1 & t & \dots & \frac{t^{k-2}}{(k-2)!} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \dots & 1 \end{bmatrix}$

$$\vec{y}(t) = P \cdot e^{Jt} = [\vec{v}^{(1)} \vec{v}^{(2)} \dots \vec{v}^{(k)}] e^{\lambda t} \begin{bmatrix} 1 & t & t^2/2! & \dots & \frac{t^{k-1}}{(k-1)!} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \dots & 1 \end{bmatrix}$$

$$= P \begin{bmatrix} e^{\lambda t} & t e^{\lambda t} & \frac{t^2}{2!} e^{\lambda t} & \dots & \frac{t^{k-1}}{(k-1)!} e^{\lambda t} \\ 0 & e^{\lambda t} & t e^{\lambda t} & \dots & \frac{t^{k-2}}{(k-2)!} e^{\lambda t} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \dots & e^{\lambda t} \end{bmatrix}$$

$\Rightarrow \left\{ e^{\lambda t}, t e^{\lambda t}, \frac{t^2}{2!} e^{\lambda t}, \dots, \frac{t^{k-1}}{(k-1)!} e^{\lambda t} \right\} \rightarrow$ construct a set of fundamental solutions for higher order ODE.

OR $\left\{ \begin{matrix} e^{\lambda t} \\ t e^{\lambda t} \\ t^2 e^{\lambda t} \\ \vdots \\ t^{k-1} e^{\lambda t} \end{matrix} \right\} \rightarrow$ is a set of fundamental solutions for higher order ODE.

$\begin{matrix} y_1 & y_2 & y_3 & \dots & y_k \end{matrix}$

Remark Consider $J = \left[\begin{array}{c|cc} \lambda & 0 & 0 \\ \hline 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{array} \right] \xrightarrow[\text{blocks}]{\text{Lus}} e^{Jt} = \left[\begin{array}{c|cc} e^{\lambda t} & 0 & 0 \\ \hline 0 & e^{\lambda t} & te^{\lambda t} \\ 0 & 0 & e^{\lambda t} \end{array} \right]$

$\{e^{\lambda t}, e^{\lambda t}, te^{\lambda t}\} \Rightarrow$ brings us only two linearly independent solutions. But, we need three.
 $\swarrow$
 linearly dependent

Ex $y'' + by' + ay = 0 \quad y(0)=1 \quad y'(0)=5$ solve.

Char eq $\lambda^2 + b\lambda + a = 0$
 $(\lambda + 3)^2 = 0 \Rightarrow \lambda_1 = \lambda_2 = -3 \quad \text{alp}(-3) = 2$

$y_1 = e^{-3t} \quad y_2 = te^{-3t}$

F.C.s $y(0)=1 \Rightarrow c_1=1$

$y'(0)=5 \Rightarrow y'_p = -3c_1 e^{-3t} + c_2 e^{-3t} - 3c_2 e^{-3t} \cdot t$

$y'_p(0) = 5 = -3c_1 + c_2 \Rightarrow c_2 = 8$

Solution of I.V.P $\underbrace{y(t) = e^{-3t} + 8te^{-3t}}$

By using systems $y'' + by' + ay = 0$

$x_1 = y$
 $x_2 = y'$

$\Rightarrow \begin{cases} x_1' = x_2 \\ x_2' = -bx_2 - ax_1 \end{cases}$

$\Rightarrow \begin{bmatrix} x_1' \\ x_2' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -a & -b \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$\det(A - \lambda I) = \begin{vmatrix} -\lambda & 1 \\ -a & -\lambda - b \end{vmatrix} = \lambda^2 + b\lambda + a = 0 \Rightarrow \lambda_{1,2} = -3$

$\lambda_1 = -3$

$\begin{bmatrix} 3 & 1 \\ -9 & -3 \end{bmatrix} \rightarrow \begin{bmatrix} 3 & 1 \\ 0 & 0 \end{bmatrix}$

$3x_1 + x_2 = 0$

$\text{span} \left\{ \begin{bmatrix} 1 \\ -3 \end{bmatrix} \right\} = \ker(A + 3I)$

$\ker(A - 3I)^2 = \begin{pmatrix} 3 & 1 \\ -9 & -3 \end{pmatrix} \begin{pmatrix} 3 & 1 \\ -9 & -3 \end{pmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

$\ker(A + 3I)^2 = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$

Choose $\vec{v}_2 \in \ker(A+3I)^2$ and $\vec{v}_2 \notin \ker(A+3I)$

Choose $\vec{v}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \Rightarrow \vec{v}_1 = (A+3I)\vec{v}_2 = \begin{bmatrix} 3 & 1 \\ -9 & -3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ -9 \end{bmatrix}$

$P = [\vec{v}_1 \ \vec{v}_2] = \begin{bmatrix} 3 & 1 \\ -9 & 0 \end{bmatrix} \quad J = \begin{bmatrix} -3 & 1 \\ 0 & -3 \end{bmatrix} \quad e^{Jt} = e^{-3t} \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}$

$$x(t) = P e^{Jt} = \begin{bmatrix} 3 & 1 \\ -9 & 0 \end{bmatrix} e^{-3t} \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix} = e^{-3t} \begin{bmatrix} 3 & 3t+1 \\ -9 & -9t \end{bmatrix}$$

$$= \begin{bmatrix} 3e^{-3t} & (3t+1)e^{-3t} \\ -9e^{-3t} & -9te^{-3t} \end{bmatrix}$$

$y_1 = 3e^{-3t} \quad y_2 = (3t+1)e^{-3t} = 3te^{-3t} + e^{-3t}$

$$y_p = c_1 y_1 + c_2 y_2 = 3c_1 e^{-3t} + c_2 3te^{-3t} + c_2 e^{-3t}$$

$$= \underbrace{(3c_1 + c_2)}_{d_1} e^{-3t} + \underbrace{3c_2 t}_{d_2} e^{-3t}$$

$$= d_1 e^{-3t} + d_2 t e^{-3t} \quad d_1, d_2?$$

$d_1 = 1 \quad d_2 = 1$

Remark If $\lambda = \alpha + i\beta$ is repeated k times, then the real solutions

$$\{ e^{\alpha t} \cos \beta t, e^{\alpha t} \sin \beta t, t e^{\alpha t} \cos \beta t, t e^{\alpha t} \sin \beta t, t^2 e^{\alpha t} \cos \beta t, t^2 e^{\alpha t} \sin \beta t, \dots, t^{k-1} e^{\alpha t} \cos \beta t, t^{k-1} e^{\alpha t} \sin \beta t \}$$

Ex $y^{(4)} + 2y'' + y = 0$

Char. eq. $\lambda^4 + 2\lambda^2 + 1 = 0$

$\lambda^2 \quad \lambda^2$

$\lambda^2(\lambda^2+1) + (\lambda^2+1) = 0$

$(\lambda^2+1)(\lambda^2+1) = 0$

$\lambda = \pm i \quad \lambda = \pm i$

$\lambda = i \quad \operatorname{Re} \lambda = 0 \quad \operatorname{Im} \lambda = 1$

$\lambda_{1,2} = i \quad \alpha(p) = 2$

$\lambda_{3,4} = -i \quad \alpha(p) = 2$

$$\{ e^{0t} \cos t, e^{0t} \sin t, t e^{0t} \cos t, t e^{0t} \sin t \} = \begin{matrix} \cos t & \sin t & t \cos t & t \sin t \\ \parallel & \parallel & \parallel & \parallel \\ y_1 & y_2 & y_3 & y_4 \end{matrix}$$

$$y_p = c_1 \cos t + c_2 \sin t + c_3 t \cos t + c_4 t \sin t$$

Test to find root of a polynomial

① $x^n + a_n x^{n-1} + \dots + a_1 x + a_0 = 0$

Assume $a_0, a_1, \dots, a_n \in \mathbb{Z}$. If the root $r \in \mathbb{Z}$ then r divides a_0 ($r | a_0$)

Ex $y^{(4)} + y''' - 7y'' - y' + 6y = 0$

$r | a_0 = 6$ $r = \pm 1, \pm 2, \pm 3, \pm 6$ candidates, put them in char. eq. and if it satisfies eq. then it is a root.
 $r = 1, -1, 2, -3$ roots.

② $a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0 = 0$

Assume $a_n, a_{n-1}, \dots, a_1, a_0 \in \mathbb{Z}$ and if $r = \frac{p}{q}$ is a root then

$q | a_n$ and $p | a_0$

Ex consider $x^4 + x^3 - 7x^2 + x + 6 = 0$ $a_4 = 1$ $a_0 = 6$

$r = \frac{p}{q}$ $q | 1 \Rightarrow q = \pm 1$

$p | 6 \Rightarrow p = \pm 1, \pm 2, \pm 3, \pm 6$

$r = \frac{p}{q} = \pm 1, \pm 2, \pm 3, \pm 6 \Rightarrow r = \underbrace{1, -1, 2, -3}_{\text{roots}}$

$\underbrace{2^4}_{16} + \underbrace{2^3}_{8} - 7 \cdot \underbrace{2^2}_{4} - 2 + 6 = 0$

$\underbrace{(-2)^4}_{16} + \underbrace{(-2)^3}_{-8} - 7 \cdot \underbrace{(-2)^2}_{4} + 2 + 6 \neq 0$
 $\quad \quad \quad = 16 - 8 - 28 + 2 + 6 = -12$

SUMMARY

n th order diff. eq. with constant coef

$$y^{(n)} + a_{n-1}y^{(n-1)} + a_{n-2}y^{(n-2)} + \dots + a_1y' + a_0y = 0 \quad \text{convert to char. eq.}$$

$$L(\lambda) = \lambda^n + a_{n-1}\lambda^{n-1} + a_{n-2}\lambda^{n-2} + \dots + a_1\lambda + a_0$$

Roots of $L(\lambda)$

Case I n real, distinct roots $\lambda_1, \lambda_2, \dots, \lambda_n$ then $y_i = e^{\lambda_i t}$ are solutions $i=1, 2, \dots, n$

Case II Complex roots $\lambda = \alpha \pm i\beta$ $y_1 = e^{\alpha t} \cos \beta t$ $y_2 = e^{\alpha t} \sin \beta t$ are fundamental solutions

Case III Repeated eigenvalues

① λ real root with $\text{alg}(\lambda) = k$

$$y_1 = e^{\lambda t} \quad y_2 = t \cdot e^{\lambda t} \quad \dots \quad y_k = t^{k-1} e^{\lambda t} \quad \text{fundamental solutions.}$$

② $\lambda = \alpha \pm i\beta$ with $\text{alg}(\lambda) = k$

$$y_1 = e^{\alpha t} \cos \beta t \quad y_2 = t \cdot e^{\alpha t} \cos \beta t \quad \dots \quad y_k = t^{k-1} e^{\alpha t} \cos \beta t$$

$$\tilde{y}_1 = e^{\alpha t} \sin \beta t \quad \tilde{y}_2 = t e^{\alpha t} \sin \beta t \quad \dots \quad \tilde{y}_k = t^{k-1} e^{\alpha t} \sin \beta t$$

are fundamental solutions

EXAMPLES

① $y''' - y'' - 2y' = 0 \quad y' = x$

$$\lambda^3 - \lambda^2 - 2\lambda = 0 \quad \lambda(\lambda+1)(\lambda-2) = 0 \quad \lambda = 0 \quad \lambda = -1 \quad \lambda = 2$$

$$\text{F.S.S} = \{e^{0t}, e^{-t}, e^{2t}\} \quad y = c_1 e^{0t} + c_2 e^{-t} + c_3 e^{2t}$$

② $y''' - 6y'' + 13y' - 12y + 4y = 0$

$$\lambda^4 - 6\lambda^3 + 13\lambda^2 - 12\lambda + 4 = 0$$

Root divides 4 so candidates

$$\lambda = \pm 1, \pm 2, \pm 4$$

$$\underline{\lambda=1} \quad 1^4 - 6 \cdot 1^3 + 13 \cdot 1^2 - 12 \cdot 1 + 4 = 0$$

$$\underline{\lambda=2} \quad 2^4 - 6 \cdot 2^3 + 13 \cdot 2^2 - 12 \cdot 2 + 4 = 0$$

$$(x-1)(x-2) = x^2 - 3x + 2.$$

$$((x-1)(x-2))^2 = x^4 - 6x^3 + 13x^2 - 12x + 4$$

$$\text{F.S.S} = \{e^t, te^t, te^{2t}, e^{2t}\}$$

$$y_g = c_1 e^t + c_2 t e^t + c_3 t e^{2t} + c_4 e^{2t}.$$

$$\textcircled{3} \quad y^{(4)} - 4y''' + 14y'' - 20y' + 25y = 0$$

$$x^4 - 4x^3 + 14x^2 - 20x + 25 = 0$$

$$(x^2 + ax + b)^2 = x^4 + 2ax^3 + (a^2 + 2b)x^2 + 2abx + b^2$$

$$2a = -4 \quad a^2 + 2b = 14$$

$$a = -2 \quad b = 5.$$

$$(x^2 - 2x + 5)^2 = x^4 - 4x^3 + 14x^2 - 20x + 25.$$

$$\lambda_{1,2} = \frac{2 \pm \sqrt{4 - 4 \cdot 5}}{2} = \frac{2 \pm 4i}{2} = 1 \pm 2i$$

$$m_1 = 1 + 2i. \quad \text{F.S.S} = \{e^t \cos 2t, e^t \sin 2t, t e^t \cos 2t, t e^t \sin 2t\}$$

$$m_2 = 1 - 2i.$$

$$\textcircled{4} \quad y^{(4)} - 9y'' + 20y = 0 \Rightarrow \lambda^4 - 9\lambda^2 + 20 = 0$$

$$(x^2 - 5)(x^2 - 4) = 0$$

$$\begin{array}{cc} x & -5 \\ x^2 & -4 \end{array}$$

$$\lambda_{1,2} = \pm \sqrt{5} \quad \lambda_{1,2} = \pm 2.$$

$$\text{F.S.S} = \{e^{\sqrt{5}t}, e^{-\sqrt{5}t}, e^{2t}, e^{-2t}\}.$$

$$y(t) = c_1 e^{\sqrt{5}t} + c_2 e^{-\sqrt{5}t} + c_3 e^{2t} + c_4 e^{-2t}.$$

5) $y^{(4)} - 2y''' + 2y'' - 2y' + y = 0$

$\lambda^4 - 2\lambda^3 + 2\lambda^2 - 2\lambda + 1 = 0$ put $\lambda = 1$ $1^4 - 2 \cdot 1^3 + 2 \cdot 1^2 - 2 \cdot 1 + 1 = 0$

$$\begin{array}{r} \lambda^4 - 2\lambda^3 + 2\lambda^2 - 2\lambda + 1 \quad \lambda - 1 \\ \underline{\lambda^4 - \lambda^3} \\ \lambda^3 - \lambda^2 + \lambda - 1 \end{array}$$

$-\lambda^3 + 2\lambda^2 - 2\lambda + 1$

$-\lambda^3 + \lambda^2$

$\lambda^2 - 2\lambda + 1$

$\lambda^2 - \lambda$

$-\lambda + 1$

$-\lambda + 1$

0

$\lambda^4 - 2\lambda^3 + 2\lambda^2 - 2\lambda + 1 = (\lambda - 1)(\lambda^3 - \lambda^2 + \lambda - 1)$

$= (\lambda - 1)(\lambda^2(\lambda - 1) + (\lambda - 1))$

$= (\lambda - 1)^2(\lambda^2 + 1)$

$\lambda = 1, 1$

$\lambda = \pm i$

F.S.S = $\{e^t, t e^t, \cos t, \sin t\}$

$y(t) = C_1 e^t + C_2 e^t t + C_3 \cos t + C_4 \sin t$

6) $y^{(4)} - y = 0$

$\lambda^4 - 1 = 0$

$(\lambda^2 - 1)(\lambda^2 + 1) = 0$

$\lambda = \pm 1$

$\lambda = \pm i$

F.S.S = $\{e^t, e^{-t}, \cos t, \sin t\}$

$y_p(t) = C_1 e^t + C_2 e^{-t} + C_3 \cos t + C_4 \sin t$

7) $y^{(5)} - 3y^{(4)} + 3y^{(3)} - y'' = 0$

$\lambda^5 - 3\lambda^4 + 3\lambda^3 - \lambda^2 = 0$

$\lambda^2(\lambda^3 - 3\lambda^2 + 3\lambda - 1) = 0$

$\lambda^2(\lambda - 1)^3 = 0$

$\lambda_1 = 0$

$\lambda_2 = 0$

$\lambda = 1, 1, 1$

F.S.S = $\{e^{0t}, t e^{0t}, e^t, t e^t, t^2 e^t\}$

$y_p(t) = C_1 + C_2 t + (C_3 + C_4 t + C_5 t^2) e^t$

8) $y^{(4)} + 4y^{(3)} + 8y'' + 8y' + 4y = 0$

$\lambda^4 + 4\lambda^3 + 8\lambda^2 + 8\lambda + 4 = 0$

$(\lambda^2 + a\lambda + b)^2 = \lambda^4 + 2a\lambda^3 + (a^2 + 2b)\lambda^2 + 2ab\lambda + b^2$

$a = 2, b = 2$

$(\lambda^2 + 2\lambda + 2)^2 = \lambda^4 + 4\lambda^3 + 8\lambda^2 + 8\lambda + 4$

F.S.S = $\{e^{-t} \cos t, e^{-t} \sin t, t e^{-t} \cos t, t e^{-t} \sin t\}$

$\lambda_{1,2} = \frac{-2 \pm \sqrt{4 - 4 \cdot 2}}{2} = \frac{-2 \pm 2i}{2} = -1 \pm i$

$y_8 = e^{-t} (C_1 + C_2 t) \cos t + e^{-t} \sin t (C_3 + C_4 t)$

9) $y''' + 4y'' - 5y' = 0$
 $\lambda^3 + 4\lambda^2 - 5\lambda = 0$
 $\lambda(\lambda^2 + 4\lambda - 5) = 0$
 $\lambda(\lambda + 5)(\lambda - 1) = 0$
 $\downarrow \quad \downarrow \quad \downarrow$
 $\lambda = 0 \quad \lambda = -5 \quad \lambda = 1$

F.S.S = $\{e^{0t}, e^{-5t}, e^t\}$

$y(t) = c_1 e^{0t} + c_2 e^{-5t} + c_3 e^t$

$y'(t) = c_2 e^{-5t} - 5c_3 e^{-5t}$

$y''(t) = -5c_2 e^{-5t} + 25c_3 e^{-5t}$

How many conditions

$y(0) = 4 \quad y'(0) = -7 \quad y''(0) = 23$

$y(0) = c_1 + c_2 + c_3 = 4$

$30c_3 = 30 \Rightarrow c_3 = 1$

$y'(0) = c_2 - 5c_3 = -7$

$c_2 = -2$

$y''(0) = c_2 + 25c_3 = 23$

$c_1 = 5$

10) $y''' + 3y'' + 3y' + y = 0$

$y(0) = 7 \quad y'(0) = -7 \quad y''(0) = 11$

$\lambda^3 + 3\lambda^2 + 3\lambda + 1 = 0$

$(\lambda + 1)^3 = 0$

F.S.S = $\{e^{-t}, t \cdot e^{-t}, t^2 e^{-t}\}$

$y(t) = c_1 e^{-t} + c_2 t e^{-t} + c_3 t^2 e^{-t} = (c_1 + c_2 t + c_3 t^2) e^{-t}$

$y'(t) = (c_2 + 2c_3 t) e^{-t} - (c_1 + c_2 t + c_3 t^2) e^{-t}$

$y''(t) = 2c_3 e^{-t} - (c_2 + 2c_3 t) e^{-t} - (c_2 + 2c_3 t) e^{-t} - (c_1 + c_2 t + c_3 t^2) e^{-t}$

$y(0) = c_1 = 7$

$y'(0) = c_2 - c_1 = -7 \quad c_2 = 0$

$y(t) = 7e^{-t} + 2t^2 e^{-t}$

$y''(0) = 2c_3 - c_2 - c_2 + c_1 = 11$

$2c_3 + 7 = 11 \quad c_3 = 2$

Ex Obtain the differential equations whose roots are given

① $2, 2, 2, 2, 1+i, 1+i, 1+i, 1-i, 1-i, 1-i$

$$p(\lambda) = (\lambda-2)^4 (\lambda - (1+i))^3 (\lambda - (1-i))^3 = (\lambda-2)^4 ((\lambda-1+i)(\lambda-1-i))^3$$

$$= (\lambda-2)^4 ((\lambda-1)^2 + 1)^3 = (\lambda-2)^4 (\lambda^2 - 2\lambda + 2)^3 = 0$$

$$(D-2)^4 (D^2 - 2D + 2)^3 y = 0 \text{ diff eqn}$$

$$F.S.S = \{e^{2t}, e^{2t}t, t^2e^{2t}, t^3e^{2t}, e^t \cos t, t e^t \cos t, t^2 e^t \cos t, e^t \sin t, t e^t \sin t, t^2 e^t \sin t\}$$

$$y_p(t) = (c_1 + c_2 t + c_3 t^2 + c_4 t^3) e^{2t} + (c_5 + c_6 t + c_7 t^2) e^t \cos t + (c_8 + c_9 t + c_{10} t^2) e^t \sin t$$

② $4, 4, 4, 2i, 2i, -2i, -2i$

$$p(\lambda) = (\lambda-4)^3 (\lambda-2i)^2 (\lambda+2i)^2 = (\lambda-4)^3 (\lambda^2+4)^2$$

$$(D-4)^3 (D^2+4)^2 y = 0$$

$$F.S.S = \{e^{4t}, t e^{4t}, t^2 e^{4t}, \cos 2t, \sin 2t, t \cos 2t, t \sin 2t\}$$

$$y_p = (c_1 + c_2 t + c_3 t^2) e^{4t} + (c_4 + c_5 t) \cos 2t + (c_6 + c_7 t) \sin 2t$$

Ex Obtain the differential equations whose solutions are given

$$\textcircled{1} \quad y(t) = C_1 e^t + C_2 e^{2t} + C_3 e^{3t} + C_4 e^{4t}.$$

$\lambda=1 \qquad \lambda=2 \qquad \lambda=3 \qquad \lambda=4$

$$[(D-1)(D-2)(D-3)(D-4)]y = 0.$$

$$y^{IV} - 10y''' + 37y'' - 50y' + 24y = 0$$

$$\textcircled{2} \quad y(x) = 2x^2 + x e^x \cos x - 4 \sin x$$

$$x^2: 0, 0, 0$$

$$\sin x: 1, 0$$

$$x e^x \cos x: (1+p)^2, (1-p)$$

$$[D^3 [D - (1+p)] [D - (1-p)]]^2 (D+p)(D-p)]y = 0$$

$$D^3 [(D-1)^2 + 1]^2 (D^2 + 1)y = 0$$

$$[D^3 ((D-1)^2 + 1)^2 (D^2 + 1)]y = 0$$

$$y_p(t) = C_1 + C_2 t + C_3 t^2 + e^t \{ (C_4 + C_5 t) \cos t + (C_6 + C_7 t) \sin t \} + C_8 \cos t + C_9 \sin t.$$